

Lab 3: Architecture Justification

EV charging station reservation system (Problem statement 21)

Architecture Selection: We chose **Microservices Architecture** for the EV Charging Station Reservation System.

Architectural Choice

The system is split into five independently deployable services behind an API Gateway component: Reservation Service, Search Service, Pricing Service, Payment Service and Station Management. Layered was rejected because scaling one layer means scaling everything. Client-Server was rejected because a single central server is a bottleneck and single point of failure during peak hours.

Two Reasons

- 1. Uneven load and NFR002.** Search is read-heavy, while reservation is write-heavy and needs strict locking to avoid double booking (FR001). Separate services let us scale Search and Reservation independently to reach 500 concurrent requests with under 2 s average response time.
- 2. Fault isolation for external and real-time parts.** The Payment Gateway is external and can fail (alternate flow 8a). In a separate Payment Service, a failure or retry only affects payment, while drivers can still search and view stations. Operator status changes (the real-time charger status requirement) live in the Station Management component and reach Search within 5 seconds.

Security Advantage

Payment card handling is isolated in the Payment Service, so card data never reaches other services and the compliance scope stays small. The API Gateway is the only public entry point: it authenticates every driver and operator request and applies rate limiting, while internal services accept traffic only from the gateway. This supports the security part of NFR001.

Performance Benefit

Each service scales horizontally on its own. Search results can be cached and served by many replicas, while the Reservation Service runs extra instances only when booking demand spikes (holidays, rush hours). Independent scaling and short, focused calls keep response time under 2 seconds without over-provisioning the whole system. The trade-off is added network latency and operational complexity, which we accept for the scalability gain.